

STUDYFLOW A WEB-BASED STUDY PLANNER FOR COURSE AND TASK MANAGEMENT

1. PROBLEM IDEA

- **Overview:** Help students organize courses, manage tasks and deadlines, track study progress, and determine which tasks should be completed first.
- **More important factors differentiate StudyFlow from a traditional to-do list:**
 - Define deadline, importance level, current progress, and optionally estimated duration to calculate task priority, and recommend what should be done next.
(*bonus: If Estimated Duration is not provided, the system uses a default duration of 3 hours*)

2. CORE FEATURES (3 standard features)

2.1. Course Management (Actor: Student)

- Create
- Edit
- View
- Delete
- Assign color
- View tasks belonging to each course

2.2. Task – Deadline Management (Actor: Student)

- Create, Edit, Delete
- Set deadlines
- Update task progress
- Set Estimate Duration (*optional*)
- Specify importance
- Mark completed
- View overdue tasks
- Sort and filter

2.3. Task Progress & Completion Tracking (Actor: Student/System)

- Track current progress of each task
- Display completed and remaining tasks
- Show task progress visually
- Automatically update task status based on progress

Authentication – Supporting Functionality

- Register
- Login/Logout

3. SMART FEATURES (Actor: System)

3.1. Smart Task Prioritization: differentiates StudyFlow from a traditional to-do list app

User provides	deadline	importance level
	current progress	estimated duration (optional)
System derives	urgency score	overdue status
	remaining workload	workload score (if Estimated Duration is available)
	<i>bonus: effective duration = estimated duration if provided</i>	
	<i>= 3h if not provided</i>	

to calculate Priority Score (0 – 100), then automatically rank pending tasks based on it, and Overdue Status will not be used to calculate P.Score. Overdue status is used to label tasks, filter, and display warning/state

All tasks use the same Priority formula, including tasks without a user-defined Estimated Duration. This allows Priority Scores to be compared consistently across the recommendation list

$$\text{Priority Score} = 50\% \text{ Urgency} + 30\% \text{ Importance} + 20\% \text{ Workload}$$

Outputs: Priority Score
Automatic Task Ranking
Recommended Next Task
Workload Warning

- Urgency score = time until deadline (user input “Deadline”)

Time until deadline	Urgency Score
> 7 days	20
4 – 7 days	40
2 – 3 days	60
1 day	80
Today	90
Overdue	100

- Importance score based on Importance level

Importance level	Importance Score
Very low	20
Low	40
Medium	60
High	80
Very high	100

- Workload score: calculated by Remaining workload

Remaining Workload	Workload Score
≤ 1	20
>1 – ≤ 2h	40
>2 – ≤ 4h	60
>4 – ≤ 6h	80
> 6h	100

With formula: $Remaining\ Workload = Effective\ Duration \times (1 - \frac{Current\ Progress}{100})$

Current Progress	Current Progress Mark
Not started	0
Started	25
Halfway	50
Almost done	75
Completed	100

Explanation for the Optional Estimated Duration: Estimated Duration is optional because users may not always be able to accurately estimate how long a task will take. If no duration is provided, StudyFlow uses a neutral default duration of **3 hours**. This default allows all tasks to use the same Priority formula and remain comparable in both local and global task rankings. The default value is only a fallback estimate and does not represent the actual duration of the task.

→ Future work: in Mobile version: study timer and actual study duration to provide more accurate workload estimation and reduce reliance on manually estimated duration.

Smart Recommendation (both local in each Course and Global ranking)

After calculating Priority Score, system sort: Priority Score DESC

Eg:

Finish DL Lab

Deadline: Tomorrow

Urgency = 80

Importance: High

Importance = 80

Estimated Duration = 6h

Progress = 50%

Remaining Workload = 3h

Workload = 60

Priority: $0.5(80) + 0.3(80) + 0.2(60) = 76$

→ Dashboard

🔥 Finish DL Lab

Priority: 76

Due tomorrow

~3h remaining

Functions: getEffectiveDuration(), calculatePriority(), rankTasks(), getRecommendedTasks()

Tie-breaking rule:

Eg:	Task	Priority Score
	A	72
	B	72

System needs to know which task should be done first

Suggested rule:

1. Priority Score DESC
2. Deadline ASC
3. Importance DESC
4. Created Time ASC

Workload Warning – Which task is becoming risky because it still requires a lot of work and its deadline is approaching

Input: reuse Smart Task Prioritization data: Deadline, Estimated Duration, Current Progress

⇒ System gets: Urgency Score, Remaining Workload → Workload Score

When show Workload Warning?

IF task is COMPLETED → no warning

ELSE IF task is OVERDUE → show OVERDUE

ELSE IF Estimated Duration is not provided by user → cannot calculate Workload Warning

ELSE IF $(U \geq 80 \text{ AND } W \geq 60) \text{ OR } (U \geq 60 \text{ AND } W \geq 80)$

→ **show WORKLOAD WARNING**

Workload Warning detects tasks that still have a large amount of work remaining while their deadlines are approaching. Unlike Priority Ranking, which determines what should be done first, Workload Warning tells the user which tasks are becoming risky because of the combination of workload and urgency.

Functions: calculateRemainingWorkload(), getWorkloadScore(), hasWorkloadWarning()

4. FUTURE WORK – FOR MOBILE APP DEVELOPMENT COURSE VERSION

- Study Timer and Actual Study Duration
- Smart Study Scheduling
- Mobile Notifications and Reminders
- Subtask management → Automatic Progress Tracking
- Personalized Workload Estimation (use historical study data to improve Estimated Duration when sufficient user history becomes available)